

SOFTWARE REQUIREMENTS SPECIFICATION

ClearSign AI

Understand Before You Sign

Task 1: Team Selection

The following team members have collaborated on this assignment:

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Task 2: Software Requirements Document

1. Preface

This Software Requirements Specification (SRS) document describes the requirements for ClearSign AI, an AI-powered legal and financial document risk analysis platform. The document has been prepared in accordance with the structure outlined in Figure 4.17 of "Software Engineering" by Ian Sommerville, 10th Edition.

Version History:

Version	Date	Authors	Description
1.0	8th March 2026	Syed Zain Ali, Mubeen Butt	Initial release of SRS document for Assignment 2

This document is intended for the following audiences: course instructors evaluating the requirements engineering process, the development team implementing the system, and stakeholders who will use or be affected by the ClearSign AI platform. The document is subject to revision as the project progresses and requirements are refined.

2. Introduction

2a. Purpose

The purpose of ClearSign AI is to democratize access to document risk analysis by providing an intelligent, AI-powered platform that enables non-expert users to understand, interpret, and assess risk in complex legal and financial documents without requiring expensive professional consultations.

Millions of freelancers, small business owners, tenants, patients, and loan applicants are required to sign legally binding documents daily. The majority lack the legal or financial literacy to identify predatory clauses, hidden obligations, or risky terms buried in dense jargon. ClearSign AI addresses this gap by leveraging Natural Language Processing (NLP) and machine learning models to automatically detect, highlight, and explain risky content in plain English.

2b. Scope

ClearSign AI is a multi-module web application accessible via a Gradio-based user interface. The system encompasses five core functional modules:

Module	Document Type	Core ML Function
Legal Contract Scanner	Employment, freelance, service contracts	Red flag clause detection and classification
Privacy Policy Auditor	Terms of Service, Privacy Policies	Data risk summarization and scoring (A–F)
Loan Agreement Analyzer	Loan offers, credit agreements	Hidden cost extraction and benchmark comparison
Medical Consent Simplifier	Surgical/treatment consent forms	Plain-English translation of medical-legal jargon
Commercial Lease QA Chat	Commercial and residential leases	Conversational question-answering on lease terms

The system is designed to operate as a standalone web application with no user authentication required for the Gradio demo. The scope excludes formal legal advice, jurisdiction-specific compliance verification, and document generation. The platform is intended as a decision-support tool, not a replacement for qualified legal or financial counsel.

3. Glossary

Term	Definition
NLP	Natural Language Processing — a branch of AI that enables machines to understand, interpret, and generate human language.

LLM	Large Language Model — a deep learning model trained on vast text corpora, capable of text generation, summarization, and Q&A (e.g., Gemini, GPT-4).
Red Flag Clause	A contractual provision that may be disadvantageous, deceptive, or legally risky to the signing party.
Non-compete Clause	A clause restricting a party from working in a competing business after the contract ends.
Indemnification Clause	A clause requiring one party to compensate the other for losses, damages, or legal costs.
Triple Net (NNN)	A commercial lease structure where the tenant pays all property expenses in addition to rent.
Benford's Law	A statistical principle stating that leading digits in many real-world datasets follow a predictable distribution; deviations may indicate fraud.
Privacy Score	A letter grade (A through F) assigned by the system to indicate the risk level of a privacy policy's data collection practices.
Balloon Payment	A large lump-sum payment due at the end of a loan term, often obscured in loan agreements.
Gradio	An open-source Python library for building interactive ML demos and web interfaces.
QA Model	Question Answering model — an NLP model that extracts answers to natural language questions from a given text context.
Streamlit	An open-source Python framework for building data science and machine learning web applications.

4. User Requirements Definition

4a. Functional Requirements

The following functional requirements describe what the system shall do from the perspective of the end user:

1. The system shall allow users to paste raw text or upload PDF documents for analysis across all five modules.
2. The system shall highlight clauses classified as 'High Risk', 'Medium Risk', or 'Low Risk' with corresponding color indicators.
3. The system shall provide a brief plain-English explanation for each identified risk clause.
4. The system shall generate a Privacy Score (A through F) for any submitted privacy policy or Terms of Service document.
5. The system shall extract and display all identified financial terms (interest rates, fees, penalties) from loan agreements in a structured table.
6. The system shall translate complex medical-legal jargon in consent forms into simple, accessible language at approximately a 6th-grade reading level.
7. The system shall allow users to ask natural language questions about an uploaded commercial lease and receive accurate, context-based answers.

8. The system shall allow users to switch between modules via a tabbed interface without reloading the application.

4b. Non-Functional Requirements

The following non-functional requirements define the quality attributes of the system:

9. The system shall return analysis results within 15 seconds of document submission under normal network conditions.
10. The system shall maintain user privacy by not storing uploaded documents beyond the active session.
11. The system shall be accessible via modern web browsers without requiring software installation.
12. The system's interface shall be intuitive enough for users with no technical background to operate without a user manual.
13. The system shall support documents of up to 50 pages or 100,000 characters in length.
14. The system shall display clear error messages when documents are unreadable, unsupported, or exceed size limits.

5. System Requirements Definition

5a. Functional Requirements

The following system-level functional requirements define the detailed behavior of the ClearSign AI platform:

Module 1 — Legal Contract Scanner:

- The system shall tokenize submitted contract text and pass it to a text classification NLP model.
- The system shall classify each clause into one of the following categories: Non-Compete, Indemnification, Limitation of Liability, Arbitration Clause, IP Assignment, Auto-Renewal, Termination Without Cause, or Other.
- The system shall assign a risk level (High / Medium / Low) to each detected clause based on the classification output.
- The system shall generate a one-to-two sentence explanation for each flagged clause using an LLM API.

Module 2 — Privacy Policy Auditor:

- The system shall parse the submitted ToS or privacy policy text and extract sections related to data collection, third-party sharing, user profiling, and account deletion.
- The system shall compute a Privacy Score from A (safest) to F (most invasive) based on the presence and severity of identified data risk categories.
- The system shall output a bullet-point summary of the most aggressive data-harvesting clauses identified.

Module 3 — Loan Agreement Analyzer:

- The system shall extract specific financial variables including APR, late payment fees, prepayment penalties, and balloon payment amounts using an information extraction model.
- The system shall compare extracted figures against predefined regulatory benchmark ranges and flag values that exceed fair-market thresholds.
- The system shall present extracted financial data in a structured comparison table with a 'Risk' column indicating normal, elevated, or predatory status.

Module 4 — Medical Consent Simplifier:

- The system shall accept complex medical-legal text and pass it to a text simplification model (T5 or LLM-based).
- The system shall output a simplified version labelled 'What This Means for You' alongside the original text.
- The system shall preserve the factual accuracy of medical information during simplification.

Module 5 — Commercial Lease QA Chat:

- The system shall accept uploaded lease documents (PDF or text) and index their content into a context window for QA retrieval.
- The system shall accept natural language questions from the user and return accurate, cited answers extracted from the uploaded document.
- The system shall indicate when a question cannot be answered from the document context and prompt the user to consult a legal professional.

5b. Non-Functional Requirements

15. Performance: The system shall process documents of up to 10,000 words within 15 seconds using standard cloud inference APIs.
16. Scalability: The system architecture shall support horizontal scaling to accommodate up to 10,000 concurrent users in a production deployment.
17. Reliability: The system shall achieve 99.5% uptime in production and gracefully degrade to partial functionality if one module fails.
18. Security: The system shall not persist uploaded documents, extracted text, or user queries beyond the active browser session.
19. Accuracy: Risk classification accuracy shall exceed 85% on a held-out evaluation dataset of labeled legal clauses.
20. Portability: The system shall operate on any device with a modern web browser and an active internet connection.
21. Maintainability: All ML model integrations shall be modular and replaceable without requiring changes to the user interface layer.
22. Compliance: The system shall display a disclaimer on every module stating that its output does not constitute legal or financial advice.

6. System Models — Use Case Definitions

The following section provides detailed use cases for the primary interactions with ClearSign AI. Use case diagrams are omitted as per assignment instructions.

Use Case 1: Scan Legal Contract for Red Flag Clauses

Field	Details
Use Case ID	UC-01
Use Case Name	Scan Legal Contract for Red Flag Clauses
Actor	Freelancer / Small Business Owner
Preconditions	User has the contract text available (as PDF or plain text). The system is running and Module 1 (Legal Contract Scanner) is active.
Main Flow	1. User navigates to the 'Legal Contract Scanner' tab.2. User pastes contract text or uploads a PDF.3. User clicks the 'Scan Contract' button.4. System tokenizes and classifies each clause using the NLP model.5. System assigns risk levels (High / Medium / Low) to each clause.6. System generates plain-English explanations for each flagged clause.7. System displays results with color-coded highlights and explanations.
Postconditions	User receives a full risk report of the contract with explanations for each flagged clause.
Alternative Flows	A1: If the uploaded file is not readable (corrupted PDF), the system displays an error message: 'Unable to extract text. Please paste the text manually.' A2: If no risky clauses are found, the system displays: 'No high-risk clauses detected. Please still consult a legal professional for critical contracts.'
Exceptions	If the NLP model API is unavailable, the system displays a service unavailability notice and prompts the user to retry.
Business Rules	The system must always display the legal disclaimer: 'This analysis does not constitute legal advice.'

Use Case 2: Audit a Privacy Policy

Field	Details
Use Case ID	UC-02
Use Case Name	Audit a Privacy Policy and Generate Privacy Score
Actor	General Internet User / Consumer
Preconditions	User has access to a Terms of Service or Privacy Policy document. The Privacy Policy Auditor module is active.
Main Flow	1. User navigates to the 'Privacy Policy Auditor' tab.2. User pastes the raw text of a privacy policy or Terms of Service.3. User clicks 'Audit Policy'.4. System parses the text and identifies data collection, third-party sharing, user profiling, and deletion policy sections.5. System computes and displays a Privacy Score from A to F.6. System outputs a bullet-point list of the most aggressive data-harvesting clauses found.

Postconditions	User understands the privacy risk level of the policy and which specific clauses to be concerned about.
Alternative Flows	A1: If the text submitted is not a privacy policy (e.g., a recipe), the system informs the user: 'This does not appear to be a privacy policy. Please submit an appropriate document.' A2: If the policy is very short (under 100 words), the system warns: 'Document may be incomplete. Results may be inaccurate.'
Exceptions	If the summarization model fails, system shows degraded output with raw extraction only, no score.
Business Rules	Privacy Score must always be accompanied by a brief explanation of what each grade means.

Use Case 3: Analyze a Loan Agreement for Hidden Costs

Field	Details
Use Case ID	UC-03
Use Case Name	Analyze Loan Agreement for Hidden Costs
Actor	Loan Applicant / Borrower
Preconditions	User has a loan agreement document (PDF or text). The Loan Agreement Analyzer module is active.
Main Flow	1. User navigates to the 'Loan Agreement Analyzer' tab. 2. User uploads a PDF or pastes the loan agreement text. 3. User clicks 'Analyze Loan'. 4. System uses an information extraction model to identify financial variables: APR, late fees, prepayment penalties, balloon payments. 5. System compares extracted values against predefined benchmark ranges. 6. System displays a structured table with each financial variable, its extracted value, the benchmark range, and a risk flag (Normal / Elevated / Predatory).
Postconditions	User has a clear, structured view of all financial obligations and can identify potentially predatory terms before signing.
Alternative Flows	A1: If no financial figures are found, the system states: 'Could not extract financial terms. Please ensure you have submitted a valid loan agreement.' A2: If extracted values fall within normal benchmarks, system displays a green 'All terms appear standard' banner.
Exceptions	If benchmark data service is unavailable, system displays extracted values without comparison flags.
Business Rules	The system must always include the disclaimer: 'Benchmark figures are based on general market data and may not reflect current regulatory limits in your jurisdiction.'

Use Case 4: Simplify a Medical Consent Form

Field	Details
Use Case ID	UC-04

Use Case Name	Simplify Medical Consent Form Language
Actor	Patient / Patient's Family Member
Preconditions	User has a medical consent form to be simplified. The Medical Consent Simplifier module is active.
Main Flow	1. User navigates to the 'Medical Consent Simplifier' tab.2. User pastes the complex medical-legal paragraph(s) into the text input area.3. User clicks 'Simplify'.4. System sends the input to the text simplification model (LLM API).5. System returns a side-by-side or stacked view: original text on one side, simplified 'What This Means for You' on the other.6. The simplified version uses accessible language at a 6th-grade reading level.
Postconditions	User understands what they are consenting to in the medical document before signing.
Alternative Flows	A1: If the submitted text is already simple (low Flesch-Kincaid score), the system notes: 'This text is already written clearly. No simplification needed.'A2: If the text contains specialized drug names or procedures, the system flags them: 'The following terms may require clarification from your medical provider.'
Exceptions	If the LLM API call times out, system displays an error and suggests the user try again in a few seconds.
Business Rules	The system shall always display: 'This simplification is for informational purposes only. Consult your doctor before signing any medical consent form.'

Use Case 5: Chat with Commercial Lease Document

Field	Details
Use Case ID	UC-05
Use Case Name	Ask Questions About a Commercial Lease
Actor	Small Business Owner / Tenant
Preconditions	User has a commercial lease document available. The Commercial Lease QA Chat module is active.
Main Flow	1. User navigates to the 'Lease QA Chat' tab.2. User uploads their commercial lease document (PDF or text).3. System processes and indexes the document content.4. User types a natural language question (e.g., 'Who is responsible for repairs?').5. User clicks 'Ask'.6. System retrieves the relevant document section using the QA model.7. System returns an answer with a cited excerpt from the lease for verification.
Postconditions	User gains a clear understanding of their specific lease obligations without needing to read the entire document.
Alternative Flows	A1: If the question cannot be answered from the document, the system responds: 'This question could not be answered from the document. Please consult a commercial property lawyer.'A2: If the user asks multiple questions, the system maintains the document context throughout the session without re-uploading.

Exceptions	If the uploaded PDF cannot be parsed (scanned image without OCR), system prompts the user to paste the text manually.
Business Rules	All answers must include a reference to the specific lease section from which the answer was derived.

Task 3: Requirements Elicitation and Analysis

Elicitation Technique

The primary elicitation technique employed for ClearSign AI was Analysis of Existing Systems combined with Brainstorming. This combined approach was selected for the following reasons:

Analysis of Existing Systems:

We conducted a systematic review of existing tools in the legal-tech and document analysis space, including DoNotPay, LegalZoom, Kira Systems, and Dext (for receipt processing). This analysis revealed significant gaps: existing solutions are either prohibitively expensive for individual users, require professional subscriptions, or address only one document type. By studying these systems, we were able to identify which features provide the most value and which pain points remain unaddressed.

Brainstorming:

The team conducted structured brainstorming sessions to generate and prioritize use cases. We mapped each document type (contracts, privacy policies, loan agreements, medical consent forms, commercial leases) to a specific user persona experiencing a distinct pain point. This technique was especially appropriate for an AI use case because it allowed us to identify which NLP tasks (classification, summarization, extraction, simplification, QA) best serve each scenario without over-engineering the solution.

Why This Approach Was Appropriate for an AI Use Case:

AI-powered document analysis is a domain where user needs are best understood through observing what professionals (lawyers, accountants) currently do manually. By analyzing existing systems and processes, we could identify the exact manual steps that ML models should automate, ensuring that the AI functionality directly addresses real workflow bottlenecks rather than solving hypothetical problems.

Stakeholder Identification

Stakeholder	Profile	Primary Pain Point Addressed by ML
Freelancer / Independent Contractor	Self-employed individuals signing client service agreements without legal support	NLP classifier detects non-compete, IP assignment, and indemnification clauses that could expose them to legal liability
General Consumer / App User	Everyday users agreeing to ToS and privacy policies for apps and websites	Summarization model extracts and scores the most invasive data collection practices, saving hours of reading

Loan Applicant / Borrower	Individuals and SMEs applying for credit or business loans	Information extraction model surfaces hidden fees and balloon payments that could trap borrowers in debt
Patient / Patient Family	Individuals signing medical consent forms while under emotional stress in healthcare settings	Text simplification model converts medical-legal jargon into accessible language, enabling true informed consent
Small Business Owner / Tenant	Entrepreneurs signing commercial leases for office or retail space	QA model allows targeted questioning about specific obligations (repairs, rent escalation, exit clauses) without reading 80+ page documents

The overarching pain point that unifies all five stakeholder groups is the information asymmetry between document drafters (lawyers, financial institutions, corporations) and the individuals signing those documents. ClearSign AI directly addresses this asymmetry by translating complex professional language into actionable risk intelligence accessible to everyone.

Gradio_Live_URL :- <https://be6f6275540c582d76.gradio.live/>

Collab_URL :-

<https://colab.research.google.com/drive/1lrC-909R0g6cM6UNRR-jFmnFDu1nL-o9?usp=sharing>